Section: **3.8 – COMPOSITE ASSEMBLIES**

Chapter: **3820**

Title: **CURVED STIFFENED COMPOSITE PANELS**

Revision: **0.4**

Date: **09/05/2020**

1 INTRODUCTION

1.1 OBJECT

This document proposes a stress sizing method for the composite skin and stringers of a cylindrical fuselage.

Failure modes checked:

- ✓ Skin and stringer strength, considering post-buckling corrections for longitudinal compression and diagonal tension in shear
- ✓ Local buckling of skin, however…
- ✓ Buckling below ultimate loads is permitted consistently with the post-buckling cut-off policy (PBCO, ratio between the ultimate allowable load to the local buckling onset, typically between 1.0 to 1.5)
- ✓ Column buckling of stringer (plus effective skin) and other potential global buckling failures of the stiffened panel
- ✓ Joint between stringer and skin (only in case the attachment is done by means of fasteners, cocured/bonded joints not covered by this document at the release date of current revision)

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1.2 SCOPE AND LIMITATIONS

Scope and basic limitations:

- ✓ Panels made of fibrous composite materials with some degree of curvature (cylindrical shape) stiffened with longitudinal stringers in axial direction

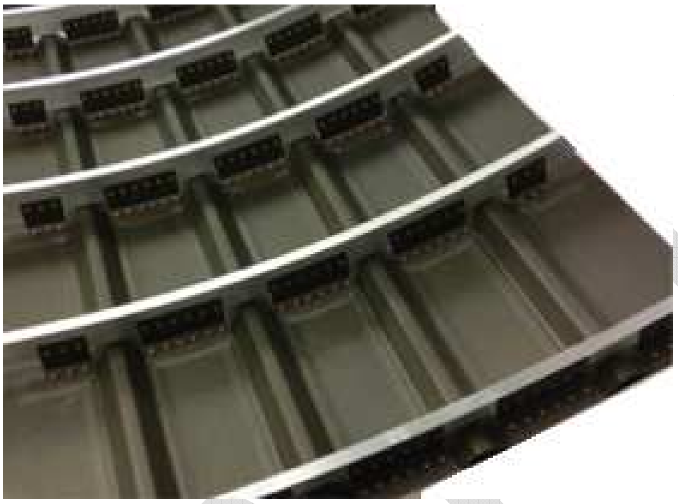

Figure 1. Curved composite panel (photo captured from website of San Diego Composites, Inc.)

- ✓ Closed form (non-numerical) methods with the minimum necessary simplification assumptions and limited accuracy
- ✓ Laminates according to classical good design rules
- ✓ Each skin subpanel is modelled as a cylindrical shell, with uniform thickness and perimeter simply supported
- ✓ The structural element considered in this analysis is the super-stringer, composed by the stringer and adjacent subpanels (halves), effectively supported at frames stations

See details in following pages…

2 STRUCTURAL DESCRIPTION

The structural configuration analysed by this method, the super-stringer, is graphically described below.

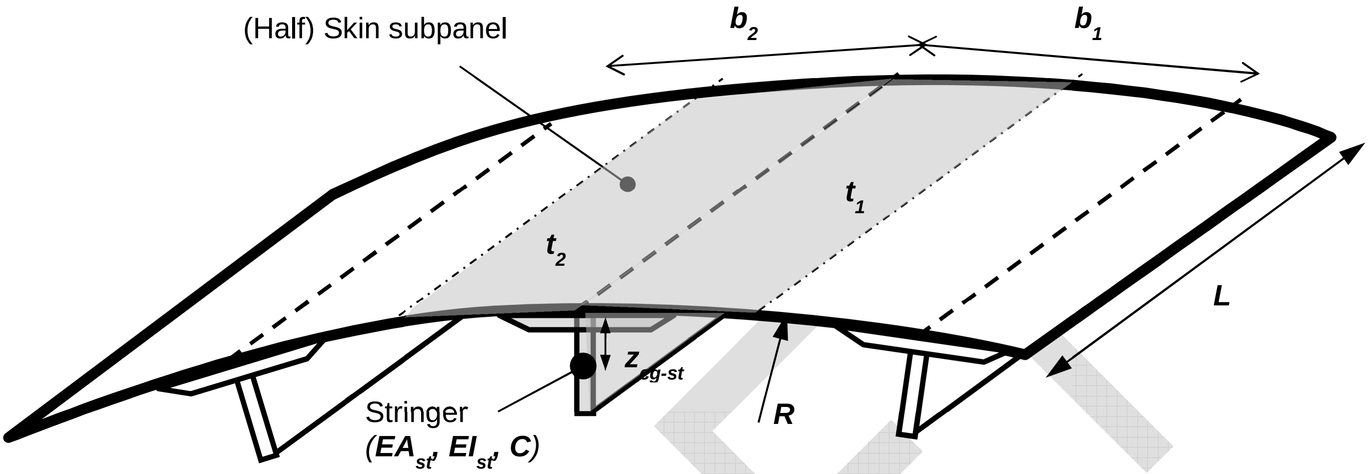

Figure 2. Super-stringer configuration, basic structural element analysed in this method

Some remarks and assumptions in this structural model and method (see details in following pages):

- Axial loads are condensed in the super-stringer cross-section forcing compatibility of longitudinal strain across the super-stringer section. On the other hand, shear and transversal loads are taken specifically for each of the two skin adjacent sub-panels.

- When analysing a complete fuselage, each skin subpanel is typically checked twice (with slightly different results), as part of two different super-stringers

- The skin subpanels enclosed among frames and stringers (L x b) are assumed simply supported at all its edges (additional torsion stiffness is conservatively ignored in local stability analyses).

- The super-stringer column is effectively supported at frames stations, with the rotation restriction controlled by means of the fixity coefficient C (for conventional designs, however, with frames including mouse-holes at stringers crossings and are not attached directly to stringers, C shall be made equal to 1.0, thus simply supported)

3 GENERAL STRESS & STRENGTH ANALYSIS PROCESS

The flow diagram below describes roughly the strength analysis process of a super-stringer location. It starts by calculating the pure-allowables, i.e. the allowable loads and strengths corresponding to the possible failure modes in uni-axial loads (laminates strengths, local buckling and crippling loads, column buckling…). Later on, for each load case, the process continues by calculating the stress/strain distribution within the different details (taking into account non-linear post-buckling effects), the corresponding failure indices and finally obtaining the minimum reserve factor (which may require additional iterations for responses that are non-linear with applied loads).

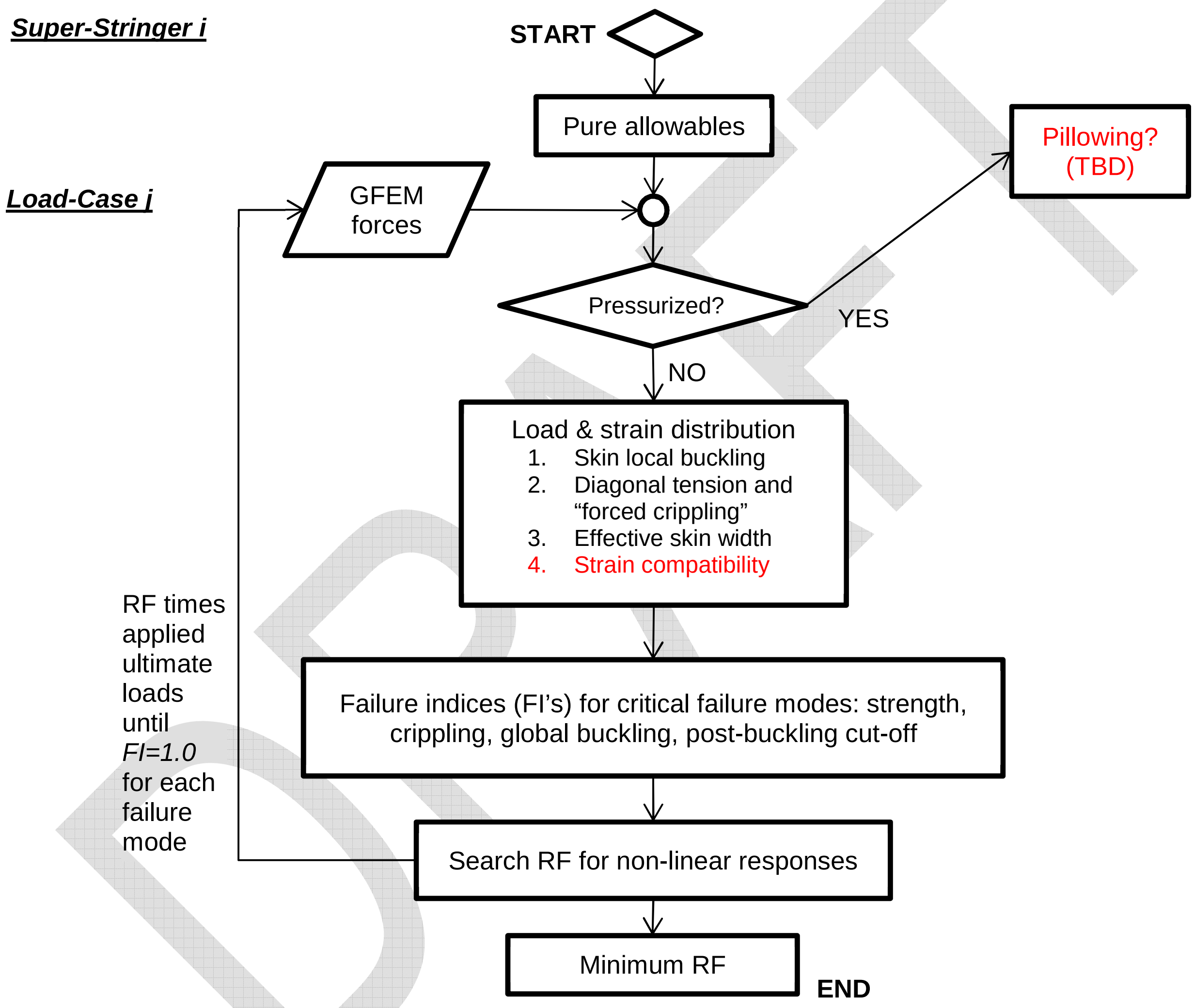

Figure 3. Flow Diagram of the standard strength check process of a super-stringer detail in a composite stiffened panel of an airframe fuselage

4 LOADS, STRESS & STRAIN

4.1 LOADS EXTRACTION FROM GFEM

Typical modelling of the fuselage skin and stringers, in a GFEM, is as follows.

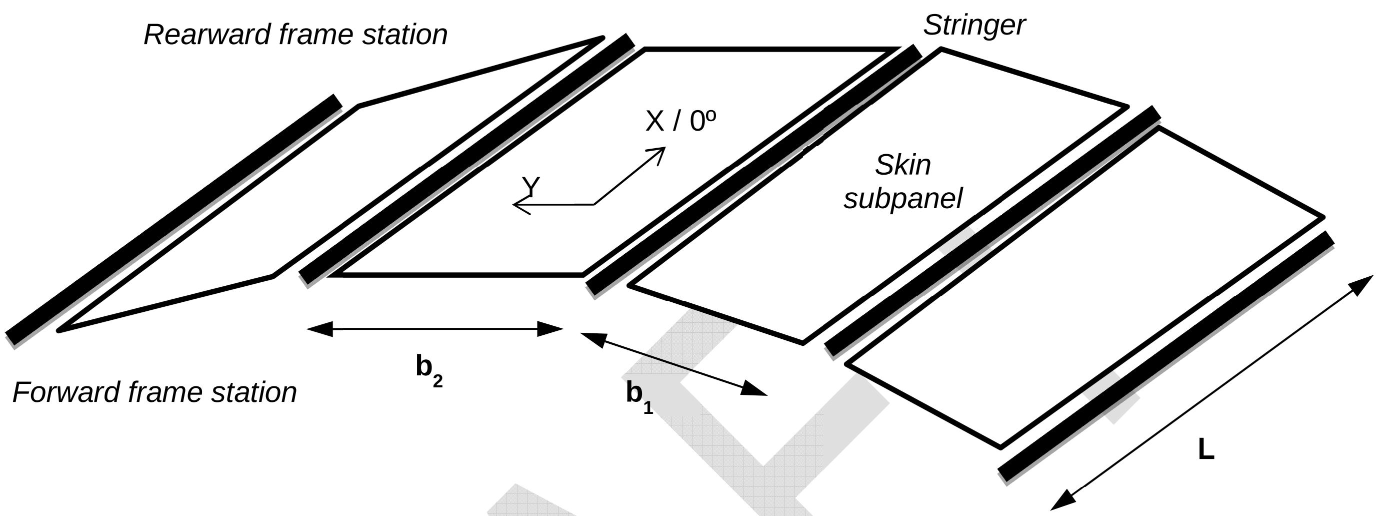

Figure 4. Stress model in GFEM

Notes:

- ✓ There may be more than one FEM element along the distance between support frames ("L" in figure above). Usually there will be only one element between stringers. Normalization approaches of internal loads for later stress and buckling stability analyses shall typically include averaging and/or taking the elements with maximum and minimum forces.
- ✓ X axis in FEM elements local reference assumed parallel to material stacking 0º direction

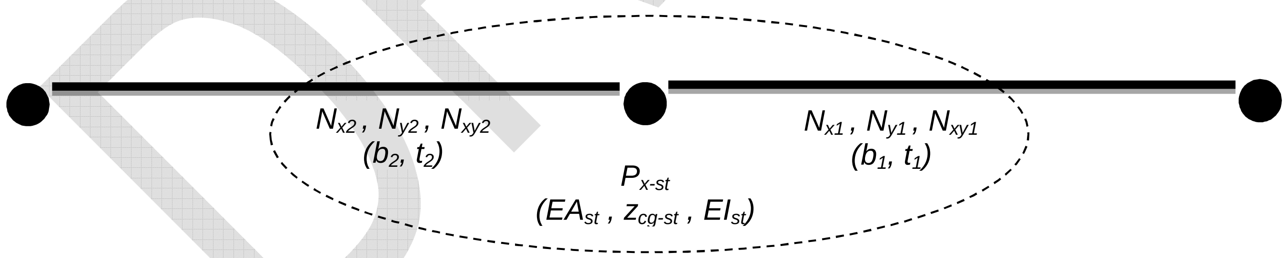

Figure 5. GFEM loads and areas of the super-stringer cross section at a certain location

Internal loads for the structural model, the super-stringer (stringer and semi-skin-subpanels):

- Axial longitudinal force: $P_x = P_{x\text{-}st} + (w_1 \cdot N_{x1} + w_2 \cdot N_{x2}) / 2$
- Axial transversal flows: N_{y1} , N_{y2}
- Shear flows: N_{xy1} , N_{xy2}

4.2 STIFFNESS OF ANALYSIS LOCATION

4.2.1 Super-stringer cross-section stiffness

Considering discussion about the effective width of skin, the centre of gravity and inertia properties of the super-stringer cross section are calculated as follows:

$$t_{max} = Max(t_1, t_2)$$

$$EA = EA_{st} + E_{x1} \cdot t_1 \cdot w_1 + E_{x2} \cdot t_2 \cdot w_2$$

$$z_{cg} = [\, EA_{st} \cdot (t_{max} + z_{cg\text{-}st}) + E_{x1} \cdot w_1 \cdot t_1^2 + E_{x2} \cdot w_2 \cdot t_2^2 \,] / EA$$

$$EI = EI_{st} + EA_{st} \cdot (t_{max} + z_{cg\text{-}st} - z_{cg})^2 + E_{x1} \cdot t_1 \cdot w_1 \cdot (t_1 - z_{cg})^2 + E_{x2} \cdot t_2 \cdot w_2 \cdot (t_2 - z_{cg})^2$$

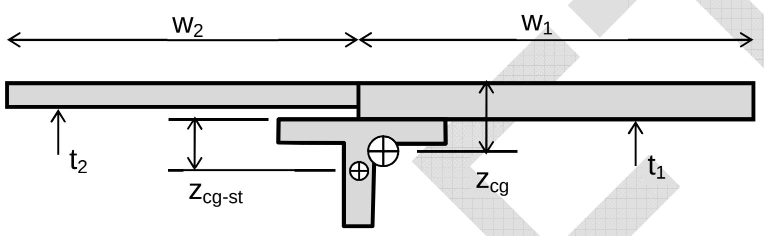

Figure 6. Super-stringer cross section

Notice the effective skin was assumed to be plane.

4.2.2 Super-stringer simplified idealization as a laminate

Idealizing the super-stringer assembly, i.e. the sum of the stringer plus the skin, as a fictitious laminate with the stiffness equivalent to the real assembly, is an approach of utility for global buckling calculations, proposed historically by several authors. References to this technique can be found, for example, in Kassapoglou's book (see Ref. 1 chapter 9.1), or, in a format developed specially for the buckling analysis of cylinders, in the NASA report SP-8007 (Ref. 2).

4.3 STRESS & STRAIN DISTRIBUTION

Strain compatibility and loads equilibrium require that:

- ✓ The axial strain ε_x shall be uniform across the section
- ✓ The membrane load flows N_y and N_{xy} shall be uniform across the strips of each subpanel (raw skin, skin pad, …)
- ✓ Contributions of the different sub-elements of the super-stringer sum the total axial loads…:
  - o total axial force $P_x = \Sigma(N_{x\text{-}i} \cdot b_i) + P_{strg}$
  - o Hook's law in stringer $P_{strg} = (EA)_{strg} \cdot \varepsilon_x$
  - o …in skin&pad laminates $[A]_i \cdot \{\varepsilon\}_i = \{N\}_i$

…with $\{\varepsilon\}_i = (\varepsilon_x, \varepsilon_{y\text{-}i}, \gamma_{xy\text{-}i})$, $\{N\}_i = (N_{x\text{-}i}, N_y, N_{xy})$

These conditions shall permit to get the stresses/strains at any point and detail of the section (note that, in the GFEM, typically details as flanges and pads will be condensed into an 1D element without the required sensitivity for some analyses)

5 FAILURE MODES

5.1 SKIN FAILURE

5.1.1 Local buckling

Forces in skin elements shall be normalized into uniform in-plane/membrane loads for each subpanel (skin bay) delimited by 2 adjacent stringers and 2 consecutive frames, which will be considered effective supports in this analysis. For typical subpanels of relatively high aspect ratios (frames spacing more than twice the stringer pitch) the recommended approach shall be averaging in width (hoop direction) and taking the elements forces values of greatest compression and shear.

Once internal loads at each subpanel have been normalized, buckling load levels shall be calculated as per chapter §3450 “PLATE/SHELL BUCKLING”.

5.1.2 Strength

Strain distribution in the super-stringer assembly is initially available per calculation method described in previous section.

Post-buckling in compression, in shear and in combined loads

For cases with axial compression, the strain distribution may need to be recalculated if local buckling of the skin is permitted below ultimate loads. Revised strain distribution shall be obtained per method in chapter §3485 “EFFECTIVE SKIN WIDTH”

For cases with skin in shear and local buckling below ultimate loads, strain distribution in skin may need to be recalculated using diagonal tension theory as per defined in chapter §3460 “SHEAR WEBS”. This approach shall force additional compression forces in stringer which shall have to be added to the internal loads calculation loop.

Skin Failure

Provided the strain state of the skin is known, taking into account post-buckling effects if any, the static margin for skin strength shall be calculated according to failure criteria set by chapter §3403 “PLAIN STRENGTH OF COMPOSITE LAMINATES” or, more typically if the part has damage tolerant requirements, those by chapter §3405 “RESIDUAL STRENGTH AFTER BVID IMPACT”.

As the response of the stress/strain with load is non-linear if local buckling is permitted (see discussion about effective skin width and diagonal tension), the reserve factor shall be preferably calculated as times the applied loads causing the failure index at skin equal to 1.0.

5.1.3 Post-buckling policy

The capacity to exceed the loads at which skin onset occurs shall be limited as a design policy, then mitigating the risks of undesired failure derived by the complexity of the post-buckling behavior (even previously defined methodology accounted post-buckling effect up to some extent). Default post-buckling policy shall be applied consistently with the proposal done by Airbus Commercial for composite fuselages (see Ref. 3).

Post-buckling-cut-off factor *PBCO* is defined as the ratio between allowable loads and the local buckling onset of the skin

$$PBCO = allowable\_load \,/\, buckling\_load$$

- ✓ If longitudinal load is tension or zero ($P_x \geq 0$): $PBCO = 1.5 / C_s$
- ✓ If shear load in panel is zero ($N_{xy} = 0$): $PBCO = 1.5 / C_c$
- ✓ If longitudinal load is compression ($P_x < 0$):

$$PBCO = PBCO_s + (PBCO_c - PBCO_s)\cdot\alpha$$

… with: $PBCO_s = 1.5 / C_s$ $PBCO_c = 1.5 / C_c$ $\alpha = 2\cdot atan(R_x/R_{xy}) / \pi$

$R_x = P_{x,b} / P_{x,b\text{-}pure}$ $R_{xy} = max[\,(N_{xy,b} / N_{xy,b\text{-}pure})_{left}\,,\,(N_{xy,b} / N_{xy,b\text{-}pure})_{right}\,]$

… being $P_{x,b}$, $N_{xy,b}$ the compression and shear components of the buckling onset in the combined-loads case, and $P_{x,b\text{-}pure}$, $N_{xy,b\text{-}pure}$ the pure buckling allowables of the skin panels under uni-axial compression and shear loads respectively,

The coefficients C_s (for shear) and C_c (for compression) used in expressions above are set function of the skin panel thickness (*t* in mm) as:

- o If $t < 2mm$ → $C_c = 1.0$ $C_s = 1.2$
- o If $2mm \leq t < 5mm$ → $C_c = 1.0 + (0.5/3)\cdot(t - 2)$ $C_s = 1.2 + (0.3/3)\cdot(t - 2)$
- o If $5mm < t$ → $C_c = 1.5$ $C_s = 1.5$

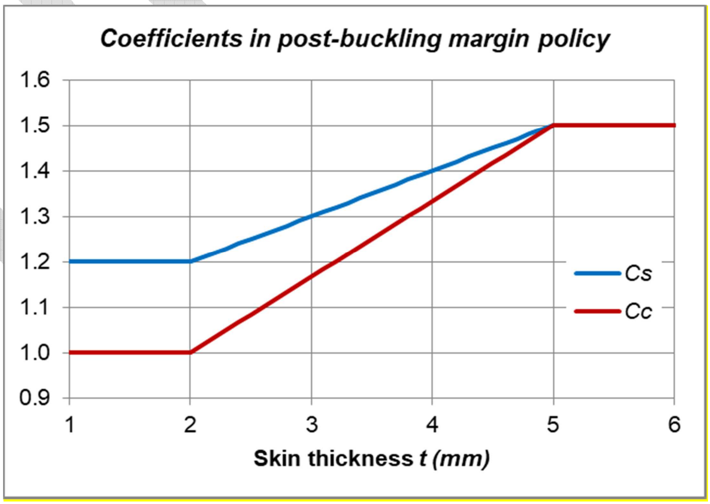

Figure 7. Post-buckling margin policy adopted as-is from Airbus Commercial (Ref. 3)

In other words, for skin thickness up to 2mm, the ultimate shear load could reach up to 1.5 times the buckling load (+50% exceedance allowed), and the ultimate axial compression up to 1.25 (1.5/1.2) times the buckling load (+25% exceedance allowed). For skin panels of high thickness no post-buckling shall be permitted.

Each project may define alternative post-buckling policies, however, provided they are adequately substantiated.

5.2 STRINGER FAILURE

5.2.1 Strength

Plain strength or residual strength after impact (per methods previously mentioned) shall be checked vs the axial strain in the stringer.

5.2.2 Local buckling and crippling

The crippling strength of the stringer shall be calculated according to method described in chapter §3430 "LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS".

As discussed for the skin, static margins for both failure modes shall be calculated preferably as times the load cause failure index equal to 1.0, taking into account non-linearities (effective skin width function of the stress).

5.3 STRENGTH OF SKIN-STRINGER JOINT

The load transfer through joint between the stringer and the skin shall normally not be a critical if standard design rules were followed, with some remarkable exceptions as stringer run-out details (out of the scope of this chapter). By-pass axial load can be critical however if the joint is riveted.

For typical super-stringer location, loads transferred through the skin-stringer joint can be extracted as explained in chapter §3215 "FORCES EXTRACTION FOR ANALYSIS OF BOLTED JOINTS". The shear load sustained by this joint shall be equal to the change in stringer axial load divided by the increment in length. With typical stringer idealizations using 1D elements:

$$N_s = \Delta P_x / \Delta L$$

5.3.1 Fastened Joint

Only in case the stringer is joined to the skin by means of rivets. The shear loaf at each rivet if attachment is done with *n* rows of rivets at pitch *p* will be:

$$Q = N_s \cdot p / n$$

Fastened joint failure

The method to calculate the static margin for the fastener joint is presented in chapter §3620 “STRENGTH OF FASTENED JOINTS IN COMPOSITES”. By-pass loads in skin and stringer foot are also required as input in this analysis, in addition to the rivet shear loads.

Inter-rivet buckling

Inter-rivet buckling stress is used as cut-out for the crippling load of the stringer in effective skin calculation. The inter-rivet buckling stress has to be obtained per method in chapter §3480 “INTER-RIVET BUCKLING”. Notice inter-rivet buckling is not failure and has not a reserve factor associated.

5.3.2 Bonded (or cocured) Joint

The shear load N_s to be sustained by this joint shall be compared with its allowable, which has to be calculated for the particular materials, laminates and geometry (foot width).

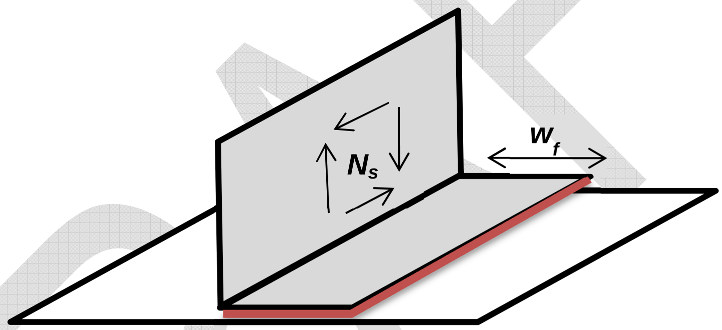

Figure 8. Sketch of a stringer foot bonding to skin

See chapters §3710 “STRESS ANALYSIS OF A BONDED JOINT” and §3730 “FAILURE CRITERION” for details about the calculation of the strength of this adhesive/cohesive attachment as in-plane-shear lap joint.

5.4 SUPER-STRINGER ASSEMBLY

5.4.1 Global Buckling in compression (Column Buckling)

The column buckling stability of the beam composed by the stringer plus the corresponding effective skin, simply supported at frames stations, under compression, shall be checked as per method defined in chapter §3440 “COLUMN BUCKLING OF COMPOSITE BEAMS”. The stringer crippling stress (and the inter-rivet buckling if applies) shall be used as cut-off value for the Johnson-Euler buckling calculation.

5.4.2 Global Buckling in combined loads (Cylinder/Barrel Buckling)

This check shall assume the frames stations are effective supports for the stiffened panel. Method described in NASA report SP-8007 (Ref. 2), shall be used for calculating the buckling onset of each fuselage slice between frames.

This analysis shall be repeated at each super-stringer location assuming the whole barrel is configured and loaded as the location of analysis.

5.4.3 Global Stability of the Fuselage

For conventional fuselages with ring frames, the frames shall provide effective support to the stiffened skin to prevent global instabilities before other local failure will occur.

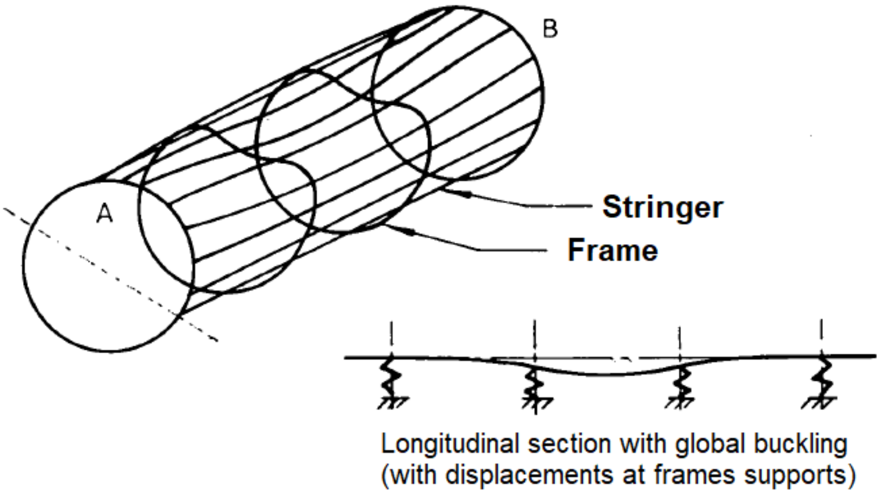

Figure 9. Fuselage general instability

A legacy method of Airbus DS used in metallic aluminium fuselage (see Ref. 4) is proposed, with a few modifications, in order to check the design of the frames is robust enough to comply with this condition. Details of the method are not shown here for being classified as Airbus DS engineering confidential.

This method is, anyway, approximated. In case the reserve factor obtained is lower than 2.5, a detailed FEM analysis of the complete fuselage shall be done.

5.5 PRESSURIZED FUSELAGES – "PILLOWING" ANALYSIS

To be completed in future revisions.

6 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
[A]	[N/mm]	In-plane stiffness matrix of the skin laminate
[B]	[N]	In-plane / bending coupling matrix of the skin laminate
b	mm	Width of the skin (distance between stringers)
b_b	mm	Buckling width of the skin, distance between nearest walls of stringers in case they are omega shaped
BL	-	Buckling Level (Buckling_Load / Applied_Load)
BVID	-	Barely Visual Impact Damage. Impact damage in the threshold of detectability.
C	-	End fixity coefficient for column buckling analysis
C_c , C_s	-	Coefficients for calculation of post-buckling margin policy
CAI	-	Compression residual strength of laminate after BVID impact, as allowable strain
[D]	[Nmm]	Bending stiffness matrix of the skin laminate
EA	N	Axial inertia of the super-stringer (stringer plus effective skin)
EA_{st}	N	Axial inertia of the stringer (without skin)
EI	Nmm^2	Bending moment of inertia of the super-stringer
EI_{st}	Nmm^2	Bending moment of inertia of the stringer (without skin)
ε	-	Strain in a composite laminate, at certain direction (typ. 0º / X , 90º / Y or ±45º)
FI	-	Failure Index, defined at detail/failure-mode level as the applied stress/strain/load divided by the allowable stress/strain/load
FRP	-	Fiber Reinforced Polymer
GFEM	-	Global Finite Element Model
l	mm	Length of the fuselage
L , a	mm	Length of the stringer, and of the adjacent skin subpanels
LL	-	Limit Load
m	-	Number of semi-waves in longitudinal X- direction of the buckling shape
n	-	Number of semi-waves in transversal Y-direction of the buckling shape, or number of rivets rows, per scope
Ns	N/mm	Shear load transferred by a longitudinal joint
N_y	N/mm	Axial transversal flow (circumferential direction)
N_{xy}	N/mm	Shear flow in skin
N_{xb}	N/mm	Buckling load of the skin subpanel in longitudinal compression
N_{yb}	N/mm	Buckling load of the skin subpanel in transversal compression (circumferential direction)
N_{xyb}	N/mm	Buckling load of the skin subpanel in shear

p	mm	Rivets pitch
P	N	Load force
PBCO	-	Ratio specifying the ultimate allowable load exceeding the buckling onset: PBCO = [Allowable_Load] / Buckling_Load
P_{col}	N	Column buckling load of the super-stringer in longitudinal compression
P_x	N	Axial longitudinal force in the whole section of the super-stringer
Q	N	Rivet shear load
R	mm	Curvature radius of the fuselage skin
RF	-	Reserve Factor (Allowable_Load / Applied_Load)
R_x	-	Failure index in X-load (X_Applied_Load / X_Allowable_Load)
R_{xy} , R_s	-	Failure index in shear XY load (XY_Applied_Load / XY_Allowable_Load)
R_y	-	Failure index in Y-load (Y_Applied_Load / Y_Allowable_Load)
R_c	-	Failure index in bi-axial loads, shear excluded (Rx+Ry if linear interaction)
strg	-	As subscript, means applicable to the stringer
t	mm	Thickness of the skin
TAI	-	Tension residual strength of laminate after BVID impact, as allowable strain
UL	-	Ultimate Load
w	mm	Effective semi-width, i.e. skin width adjacent (at one side) to the stringer support that is still sustaining compression after buckling
z_{cg}	mm	Distance from outer skin face to the super-stringer center of gravity
$z_{cg\text{-}st}$	mm	Distance from the inner skin face to the stringer center of gravity (without skin)

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[Christos Kassapoglou]	2nd Ed.	Design and Analysis of Composite Structures
Ref. 2	[NASA] SP-8007	1968	BUCKING OF THIN-WALLED CIRCULAR CYLINDERS
Ref. 3	[Airbus] RP0721288	2.0	A350 XWB Fuselage - Stiffened Panel Typical Sizing Process
Ref. 4	NT-T-ADS-93067	Ed. A	Inestabilidad General de Fuselajes

SOFTWARE

Program	*Version*	*Description*

Table 2. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	*Authors*	*Change Reason*	*Pages/Sub-chapters affected*
0.4 09/05/2020	G. Baños	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	*Getafe*	*Manching*
Stress Methods (TEASP-TL3)	G. Baños 09/05/2020	F. Glock DD/05/2020

Table 4. *Approval signatures table for last revision of this chapter*